

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

GAUSSIAN PROCESSES

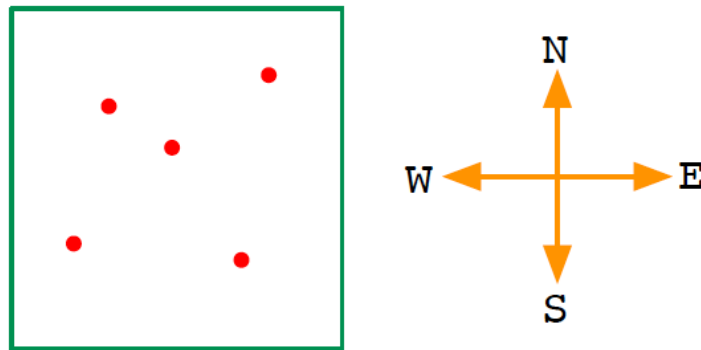
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Example: County Pollution Levels

Want a pollution map for San Diego county:



At any given time, have readings $y \in \mathbb{R}$ at a few locations $x \in [0, 1]^2$:

- Fixed county facilities
- Individual citizen reports

Use these to create a full pollution map for the county.

Given readings $(x_1, y_1), \dots, (x_n, y_n)$, infer $f: [0, 1]^2 \rightarrow \mathbb{R}$.

A Prior on Smooth Functions

Let $\mathcal{X}$ be an input space, e.g., $[0, 1]^2$.

A Gaussian in infinite dimension: a distribution over all functions $f: \mathcal{X} \rightarrow \mathbb{R}$.

- A **Gaussian process** (GP) on $\mathcal{X}$ is a collection of random variables indexed by $\mathcal{X}$ such that any finite subset of them has a Gaussian distribution.

Pick any $x_1, \dots, x_s \in \mathcal{X}$.

If f is sampled from a GP then $(f(x_1), \dots, f(x_s))$ has a Gaussian distribution.

- A Gaussian process is specified by
 - Mean function $m: \mathcal{X} \rightarrow \mathbb{R}$
 - Covariance function $k: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$
- If $f \sim GP(m, k)$ then for any finite subset $S \subset \mathcal{X}$, we have $f_S \sim N(m_S, k_{SS})$. Here f_S is a shorthand for $(f(x): x \in S)$.

Example

A distribution over functions on the line:

- $X = \mathbb{R}$
- $m(x) \equiv 0$
- $k(x, x') = \exp(-(x - x')^2)$

Sampling a Function from a GP

For $t = 1, 2, 3, \dots$:

- Pick some $x_t \in \mathcal{X}$
- The joint distribution $(f(x_1), \dots, f(x_t))$ is Gaussian
Thus so is the conditional distribution $f(x_t) | f(x_1), \dots, f(x_{t-1})$
- Sample $f(x_t)$ from this conditional distribution

Conditioning formula: for $S \subset \mathcal{X}$,

$$f(x) | f_S \sim N(m(x) + k(x, S)k(S, S)^{-1}f_S, k(x, x) - k(x, S)k(S, S)^{-1}k(S, x)).$$

Example (cont'd)

$$\mathcal{X} = \mathbb{R}, m(x) \equiv 0, k(x, x') = \exp(-\|x - x'\|^2)$$



Several Draws

